EXPERIMENT-1

1. Scenario: You are working on a project that involves analyzing student performance data for a class of 32 students. The data is stored in a NumPy array named student_scores, where each row represents a student and each column represents a different subject. The subjects are arranged in the following order: Math, Science, English, and History. Your task is to calculate the average score for each subject and identify the subject with the highest average score.

Question: How would you use NumPy arrays to calculate the average score for each subject and determine the subject with the highest average score? Assume 4x4 matrix that stores marks of each student in given order.

Aim

To calculate the average score for each subject using NumPy arrays and identify the subject with the highest average score.

Algorithm

1. Import the NumPy library.
2. Create a NumPy array containing the marks of students in Math, Science, English, and History.
3. Calculate the average score of each subject using np.mean() with axis=0.
4. Find the index of the highest average using np.argmax().
5. Display the average score of each subject.
6. Display the subject with the highest average score.

Code:

```
import numpy as np

student_scores = np.array([

  [80, 75, 85, 70],
```

```
    [90, 85, 80, 75],
    [70, 80, 75, 85],
    [85, 90, 88, 80]
])
subjects = ["Math", "Science", "English", "History"]
average_scores = np.mean(student_scores, axis=0)
highest_index = np.argmax(average_scores)
print("Average score for each subject:")
for i in range(4):
    print(subjects[i], ":", average_scores[i])
print("Subject with highest average:",
      subjects[highest_index])
```

Input:

80 75 85 70

90 85 80 75

70 80 75 85

85 90 88 80

Output:

Average score for each subject:

Math : 81.25

Science : 82.5

English : 82.0

History : 77.5

Subject with highest average: Science



Result

The average score of each subject was calculated successfully, and **Science** was identified as the subject with the highest average score.